

These are notes for Malek Abdesselam's Functional Analysis class for Fall 2026.

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Motivation

Consider a space V of (sufficiently nice) functions $f: \mathbb{R} \rightarrow \mathbb{R}$. Then, similar to how a vector in $\mathbb{R}^n$ can be denoted as

$$x = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix},$$

we may view the function $f: \mathbb{R} \rightarrow \mathbb{R}$ as a “column vector” with uncountable index. That is, $f = (f(x))_{x \in \mathbb{R}}$. If $L: V \rightarrow V$ is a linear map, then an analogue of the finite-dimensional matrix representation can be viewed as a kernel that f is integrated against; that is,

$$(Lf)(x) = \int_{\mathbb{R}} K(x, y) f(y) dy.$$

In the special case that $L = \text{id}_V$, then the kernel $K(x, y)$ is given by the Dirac delta distribution, $K(x, y) = \delta(x - y)$. The space of these distributions is the primary object of study in Fourier analysis.

Notation

- The natural numbers are given by $\mathbb{N} = \{0, 1, 2, 3, \dots\}$.
- The strictly positive natural numbers are given by $\mathbb{Z}_{>0} = \{1, 2, 3, \dots\}$.
- A multi-index is a tuple $\alpha \in \mathbb{N}^n$. We let

$$|\alpha| = \alpha_1 + \dots + \alpha_n.$$

If $x = (x_1, \dots, x_n) \in \mathbb{R}^n$, then we will write

$$x^\alpha = x_1^{\alpha_1} \dots x_n^{\alpha_n}$$

and

$$\alpha! = \alpha_1! \dots \alpha_n!$$

for the corresponding values.

If $f: \mathbb{R}^n \rightarrow \mathbb{R}$ is any function, then we will write

$$\partial^\alpha f = \frac{\partial^{|\alpha|}}{\partial x_1^{\alpha_1} \dots \partial x_n^{\alpha_n}}.$$

- If A is a set, we will write $\mathcal{P}(A)$ for the power set of A . If A and B are sets, then the set $B^A \subseteq \mathcal{P}(A \times B)$ is the set of all functions $f: A \rightarrow B$.

We will write $\mathcal{P}_{\text{fin}}(X)$ to consist of all finite subsets of X .

- If $x \in \mathbb{R}^n$ is given by $(x_1, \dots, x_n)$, we will define the bracket

$$\begin{aligned}\langle x \rangle &:= \sqrt{1 + x_1^2 + \dots + x_n^2} \\ &= \sqrt{1 + |x|^2}.\end{aligned}$$

In the particular case of $n = 1$, we will write $\langle k \rangle = \sqrt{1 + k^2}$ for any $k \in \mathbb{N}$. Note here that $|\cdot|$ denotes the norm on Euclidean space.

- If $n \in \mathbb{N}$, then

$$[n] = \{1, 2, \dots, n\},$$

where we have $[0] = \emptyset$.

- If A is a finite set, then $|A|$ is the cardinality of A .
- Generally we will denote by $\mathbb{K}$ for either $\mathbb{R}$ or $\mathbb{C}$.

Locally Convex Topological Vector Spaces

Recall that if X is a set, then $\mathcal{T} \subseteq \mathcal{P}(\mathcal{P}(X))$ is a topology on X if it satisfies:

- $\emptyset, X \in \mathcal{T}$;
- $U_1, U_2 \in \mathcal{T}$ implies $U_1 \cap U_2 \in \mathcal{T}$;
- $\{U_i : i \in I\} \subseteq \mathcal{T}$ implies $\bigcup_{i \in I} U_i \in \mathcal{T}$.

Definition: Definitions

We say $(V, \mathcal{T})$ is a *topological vector space* (over $\mathbb{K}$) if V is a $\mathbb{K}$ -vector space whose topology is such that the maps

$$\begin{aligned}V \times V &\rightarrow V \\ (u, v) &\mapsto u + v\end{aligned}$$

and

$$\begin{aligned}\mathbb{K} \times V &\rightarrow V \\ (\lambda, v) &\mapsto \lambda v\end{aligned}$$

are continuous with respect to the product topology. Note that $\mathbb{K}$ is equipped with its traditional topology.

The morphisms of topological vector spaces are the *continuous* linear maps. We say $V \cong W$ are isomorphic as topological vector spaces if there is $\varphi: V \rightarrow W$ that is bijective, linear, and continuous, and φ^{-1} is also continuous.

Definition: Let V be a $\mathbb{K}$ -vector space. We say $\rho: V \rightarrow [0, \infty)$ is a *seminorm* if

$$\begin{aligned}\rho(v + w) &\leq \rho(v) + \rho(w) \\ \rho(\lambda v) &= |\lambda| \rho(v)\end{aligned}$$

for any $v, w \in V$ and $\lambda \in \mathbb{K}$.

Remark: The only difference between a seminorm and a norm is that $\rho(v) = 0$ need not imply that $v = 0$. A norm is a seminorm.

We will let $\text{SN}(V)$ be the set of all seminorms of V .

Definition: Let V be a $\mathbb{K}$ -vector space. Let $\mathcal{N} \subseteq \text{SN}(V)$. Define $\mathcal{T}(\mathcal{N})$ to be a topology with basic open sets $U(x, F, \varepsilon) \subseteq V$ given by

$$\begin{aligned} U(x, F, \varepsilon) &= \{y \in V : \rho(y - x) < \varepsilon \text{ for all } \rho \in F\} \\ &= \bigcap_{\rho \in F} U(x, \rho, \varepsilon). \end{aligned}$$

for any $x \in V$, $F \subseteq \mathcal{N}$ finite, and $\varepsilon > 0$.

Remark: I prefer to use the subbasic open sets

$$U(x, \rho, \varepsilon) = \{y \in V : \rho(y - x) < \varepsilon\}$$

and write intersections.

We observe that the subsets of the form $U(x, \{\rho\}, \varepsilon)$ as ρ ranges across $\mathcal{N}$ forms a subbase for the topology of V .

Proposition: If $F \subseteq \mathcal{N}$ is finite, $\varepsilon > 0$, and $x \in V$, then

$$U(x, F, \varepsilon) \in \mathcal{T}(\mathcal{N}).$$

Proof. Suppose $y \in U(x, F, \varepsilon)$. Our goal is to find $\eta > 0$ such that $U(y, F, \eta) \subseteq U(x, F, \varepsilon)$. Suppose $z \in U(y, F, \eta)$. If $\rho \in F$, then

$$\begin{aligned} \rho(z - x) &\leq \rho(z - y) + \rho(y - x) \\ &< \eta + \rho(y - x). \end{aligned}$$

Therefore, we will define η to be small enough such that $\eta + \max_{\rho \in F} \rho(y - x) < \varepsilon$. This gives the desired result. $\square$

Definition: A topological space is called a locally convex topological vector space (locally convex topological vector space) if the topology is generated by a family of seminorms $\mathcal{N}$ with the property that

$$\bigcap_{\rho \in \mathcal{N}} \{v : \rho(v) = 0\} = \{0\}.$$

A family $\mathcal{N}$ of seminorms that satisfies this property is said to be *separating*.

Example: If H is a Hilbert space with orthonormal basis $\{e_n : n \in \mathbb{N}\}$, then the seminorms $v \mapsto |\langle v, e_i \rangle|$ are separating.

Theorem: If V is a $\mathbb{K}$ -locally convex topological vector space and $\mathcal{N}$ is a collection of seminorms that defines the topology on V , then the pair $(V, \mathcal{T}(\mathcal{N}))$ is Hausdorff if and only if $\mathcal{N}$ separates the points of V .

Proof. Assume $\mathcal{N}$ separates the points of V . Suppose $v \neq w$ are distinct elements of V . Then, since $v - w \neq 0$, there is some $\rho \in \mathcal{N}$ such that $\rho(v - w) = \varepsilon_0 > 0$. Consider the finite set $F = \{\rho\}$, and choose some $0 < \varepsilon < \varepsilon_0/2$. Then, we claim that the basic open sets $U(v, F, \varepsilon)$ and $U(w, F, \varepsilon)$ are disjoint. This follows from the fact that if $x \in U(v, F, \varepsilon)$, then $\varepsilon_0 = \rho(v - w) \leq \rho(v - x) + \rho(x - w)$, so that $\rho(w - x) \geq \varepsilon_0 - \varepsilon > \varepsilon_0 - \varepsilon_0/2 > \varepsilon$, yielding the desired result.

Now, suppose $\mathcal{N}$ does not separate the points of V . Then, there are some distinct $v \neq w$ such that for all $\rho \in \mathcal{N}$, we have $\rho(v - w) = 0$. In particular, observe that for all finite subsets $F \subseteq \mathcal{N}$ and all $\varepsilon > 0$, we have that w is contained in $U(v, F, \varepsilon)$, meaning that there are no basic open sets that separate v and w . Since any open subset of V contains a basic open subset, it follows that for any open subsets U and U' , we have some finite F such that $U(v, F, \varepsilon) \subseteq U$, but this also has $w \in U(v, F, \varepsilon)$, implying that $U \cap U' \neq \emptyset$. Thus, V is not Hausdorff. $\square$

Examples

Remark: If we remove the assumption that the family $\mathcal{N}$ is separating, then the topology induced by the trivial seminorm $\rho: V \rightarrow [0, \infty)$ is the indiscrete topology on V . Similarly, there is a family of seminorms that induces the discrete topology on V .

- (i) Let $(V, \|\cdot\|)$ be a normed vector space over $\mathbb{K}$. Examples of these include Banach spaces such as $L^p(X, \Sigma, \mu)$, where Σ is a σ -algebra and μ is a positive measure on the measurable space (X, Σ) , and $p \in [1, \infty]$.

There is a metric on this normed space $d(x, y) = \|x - y\|$, which gives rise to the metric topology $\mathcal{T}(d)$. We have $U \in \mathcal{T}(d)$ if and only if $U \subseteq V$ is such that for any $x \in U$, there exists $\varepsilon > 0$ such that

$$\{y \in V : d(x, y) < \varepsilon\} =: U(x, \varepsilon) \subseteq U.$$

This is a locally convex space with seminorm $\mathcal{N} = \{\|\cdot\|\}$. There are two possibilities for finite subsets $F \subseteq \{\|\cdot\|\}$, meaning that if $F = \emptyset$, then $U(x, F, \varepsilon) = U(x, \emptyset, \varepsilon) = V$, while if $F = \{\|\cdot\|\}$, then

$$\begin{aligned} U(x, F, \varepsilon) &= \{y \in V : \rho(x - y) < \varepsilon \text{ for all } \rho \in F\} \\ &= \{y \in V : \|x - y\| < \varepsilon\} \\ &= U(x, \varepsilon). \end{aligned}$$

- (ii) Define $s(\mathbb{N}, \mathbb{K})$ to be the space of sequences $x = (x_n)_{n \geq 0}$ in $\mathbb{K}^{\mathbb{N}}$ such that for all $k \geq 0$, $\sup_{n \geq 0} \langle n \rangle^k |x_n| < \infty$, where $\langle n \rangle = \sqrt{1 + n^2}$ is the bracket. This is the space of *rapidly decaying sequences* in $\mathbb{K}$. In essence, elements of $s(\mathbb{N}, \mathbb{K})$ decay faster than any inverse power of the index n .

We define the seminorms

$$\rho_k(x) = \sup_{n \geq 0} \langle n \rangle^k |x_n|.$$

In fact, the $\{\rho_k : k \in \mathbb{N}\}$ are norms. Let

$$\mathcal{N} := \{\rho_k : k \in \mathbb{N}\}$$

We obtain a locally convex on $s(\mathbb{N}, \mathbb{K})$ with topology induced by the family $\mathcal{N}$.

Later, we will see that $(s(\mathbb{N}, \mathbb{K}), \mathcal{T}(\mathcal{N}))$ is *not* a normed space. That is, there does not exist a norm $\|\cdot\|$ such that $\mathcal{T}(\mathcal{N}) = \mathcal{T}(\{\|\cdot\|\})$.

- (iii) Take $V = C([0, \infty), \mathbb{R})$ to be the set of all real-valued continuous functions on $[0, \infty)$ with pointwise addition and scalar multiplication. This is a real vector space.

At first, we may think $\|f\| = \sup_{t \in \mathbb{R}} |f(t)|$ would define a norm, but this is not satisfactory since the functions in V are not necessarily bounded.

Therefore, we want to restrict to compact subsets. Take

$$\rho_k(f) := \sup_{0 \leq t \leq k} |f(t)|.$$

Notice that the ρ_k are not norms (use Urysohn's Lemma or your favorite piecewise function to construct one). However, the family $\mathcal{N} = \{\rho_k : k \geq 0\}$ is a separating family.

Consider the set of functions $V_0 = \{f \in V : f(0) = 0\}$. Then, $V_0 \leq V$ is a closed subset.

Proof. Consider the linear functional $\text{ev}_0: V \rightarrow \mathbb{R}$ given by $f \mapsto f(0)$. We claim that this linear functional is continuous.

Let $U \subseteq \mathbb{R}$ be open. Let $z \in U$, and let $\varepsilon > 0$ be such that $U(z, \varepsilon) := \{y \in \mathbb{R} : |z - y| < \varepsilon\}$ is contained in

U . Let $f \in \text{ev}_0^{-1}(U(z, \varepsilon))$, and set $\varepsilon_0 = |z - f(0)|$. To show that $\text{ev}_0^{-1}(U(z, \varepsilon))$ is open, it suffices to find some seminorm ρ_k and some $\delta > 0$ such that $U(f, \rho_k, \delta) \subseteq \text{ev}_0^{-1}(U(z, \varepsilon))$.

In fact, we may choose $k = 1$ and $\delta = \varepsilon - \varepsilon_0$. Suppose $g \in U(f, \rho_1, \delta)$. Then, we have that

$$\begin{aligned} |g(0) - z| &\leq |g(0) - f(0)| + |f(0) - z| \\ &\leq \sup_{t \in [0, 1]} |g(t) - f(t)| + \varepsilon_0 \\ &< \delta + \varepsilon_0 \\ &= \varepsilon. \end{aligned}$$

In particular, Thus, $U(f, \rho_1, \delta) \subseteq \text{ev}_0^{-1}(U(z, \varepsilon))$, so that $\text{ev}_0^{-1}(U(z, \varepsilon))$ is open as any element in this set admits a basic open neighborhood contained in the set. Therefore, ev_0 is continuous.

Since $V_0 = \text{ev}_0^{-1}(\{0\})$, it follows that V_0 is closed since $\{0\} \subseteq \mathbb{R}$ is closed. $\square$

We claim that the subspace topology on V_0 is given by $\mathcal{T}_0 = \mathcal{T}(\{\rho_k|_{V_0} : k \geq 0\})$.

Proof. First, we will show that, given a subbasic open neighborhood $U = V_0 \cap U(x, \rho_k, \varepsilon)$ in the subspace topology of V_0 and any element y in this subbasic open neighborhood, there is an open neighborhood $V \in \mathcal{T}(\{\rho_k|_{V_0} : k \geq 0\})$ such that $y \in V \subseteq U$.

If $y \in U$, we observe that $y \in U(x, \rho_k, \varepsilon)$. In particular, since these form a subbase for $\mathcal{T}(\{\rho_k : k \geq 0\})$, it follows that there is some finite collection $F = \{\rho_{i_j} : j = 1, \dots, m\}$ of seminorms and some $\delta > 0$ such that

$$U(y, F, \delta) = \bigcap_{j=1}^m U(y, \rho_{i_j}, \delta).$$

Since $U(y, \rho_{i_j}|_{V_0}, \delta) \subseteq U(y, \rho_{i_j}, \delta)$ for each j , it follows that we have

$$\bigcap_{j=1}^m U(y, \rho_{i_j}|_{V_0}, \delta) \subseteq V_0 \cap \left(\bigcap_{j=1}^m U(y, \rho_{i_j}, \delta) \right),$$

whence setting V to equal the left hand side yields the desired result. Therefore, $\mathcal{T}(\{\rho_k|_{V_0} : k \geq 0\})$ refines the subspace topology $\mathcal{T}_0$.

In the reverse direction, observe that if $y \in U(x, \rho_j|_{V_0}, \varepsilon)$ in V_0 , then $y \in U(x, \rho_j, \varepsilon)$, so there is some $\delta > 0$ and finite $F = \{\rho_{i_j} : j = 1, \dots, m\}$ such that

$$\begin{aligned} U(y, F, \varepsilon) &= \bigcap_{j=1}^m U(y, \rho_{i_j}, \varepsilon) \\ &\subseteq U(x, \rho_j, \varepsilon), \end{aligned}$$

but we notice then that $U(x, \rho_j, \varepsilon) \cap V_0 \subseteq U(x, \rho_j|_{V_0}, \varepsilon)$, so that $V_0 \cap U(y, F, \varepsilon) \subseteq U(x, \rho_j|_{V_0}, \varepsilon)$, so that $\mathcal{T}_0$ refines $\mathcal{T}(\{\rho_k|_{V_0} : k \geq 0\})$. Thus, these topologies are equal. $\square$

We may consider $\mathcal{B}(\mathcal{T}_0)$ to be the Borel σ -algebra on V_0 . Brownian motions are a special class of probability measure on $(V_0, \mathcal{B}(\mathcal{T}_0))$. Specifically, they are the probability measures such that for every $n \geq 0$, partition $0 < t_1 < t_2 < \dots < t_n$, the random variables

$$\begin{aligned} X_1 &= f \mapsto f(t_1) \\ X_2 &= f \mapsto (f(t_2) - f(t_1)) \end{aligned}$$

$$\begin{aligned}
X_3 &= f \mapsto (f(t_3) - f(t_2)) \\
&\vdots \\
X_n &= f \mapsto (f(t_n) - f(t_{n-1}))
\end{aligned}$$

are independent centered Gaussian random variables with variances $t_1, t_2 - t_1, \dots, t_n - t_{n-1}$ for an arbitrary f .

Similar to the case of $s(\mathbb{N}, \mathbb{K})$, the space $(V_0, \mathcal{T}_0)$ is not a normed space.

- (iv) Consider the vector space $V = \mathbb{R}^{\mathbb{N}}$, which is the space of all real-valued sequences. Let $\mathcal{T}$ be the product topology on $\mathbb{R}^{\mathbb{N}}$. We are curious as to whether $(V, \mathcal{T})$ is a locally convex topological vector space.

Consider the function $d: V \times V \rightarrow [0, \infty)$ defined by

$$d(x, y) = \sum_{n=0}^{\infty} 2^{-n} \min(1, |x_n - y_n|).$$

Then, we have that d is a distance metric, and $\mathcal{T} = \mathcal{T}(d)$.

Proof. We start by showing that if $x \in V$ is any element and $\varepsilon > 0$, then $U(x, \varepsilon)$ is open in $\mathcal{T}$. Translating, we take $x = 0$ (since the metric d is translation-invariant), so that we are now seeking to show that $U(0, \varepsilon)$ is open. This is equivalent to showing that if $x \in U(0, \varepsilon)$, then there is a finite collection of subbasic open sets $\pi_j^{-1}(U_j)$ whose intersection W contains x and is contained in $U(0, \varepsilon)$.

To start, we observe that there is some index n_0 such that

$$\sum_{n=n_0+1}^{\infty} 2^{-n} < \varepsilon/2.$$

We then select $U_j = (x_j - \varepsilon/4, x_j + \varepsilon/4)$ for each $j = 0, \dots, n_0$. It then follows that $x \in W$, and we have for any $y \in W$ that

$$\begin{aligned}
d(x, y) &= \sum_{j=0}^{\infty} 2^{-j} \min(1, |x_j - y_j|) \\
&\leq \sum_{j=0}^{n_0} 2^{-j} \min(1, |x_j - y_j|) + \sum_{j=n_0+1}^{\infty} 2^{-j} \\
&< \sum_{j=0}^{n_0} 2^{-j} (\varepsilon/4) + \varepsilon/2 \\
&\leq \varepsilon,
\end{aligned}$$

whence $x \in W \subseteq U(0, \varepsilon)$.

Now, consider a basic open set $W = \bigcap_{j=0}^{n_0} \pi_j^{-1}(U_j)$. Let $x \in W$. Let $\delta_0, \dots, \delta_{n_0}$ be such that $(x_j - \delta_j, x_j + \delta_j) \subseteq U_j$ for each $j = 0, \dots, n_0$, and set $\delta = 2^{-n_1} \min(\delta_0, \dots, \delta_{n_0})$, where n_1 is any integer strictly larger than n_0 .

Notice then that if $y \in U(x, \delta)$, then we have for each $j = 0, \dots, n_0$ that

$$\begin{aligned}
|x_j - y_j| &< 2^j \delta \\
&\leq 2^{n_0} 2^{-n_1} \min(\delta_0, \dots, \delta_{n_0}) \\
&< \min(\delta_0, \dots, \delta_{n_0})
\end{aligned}$$

$$\leq \delta_j.$$

In particular, this means we must have $U(x, \delta) \subseteq W$.

Therefore, we have that $\mathcal{T}$ and $\mathcal{T}(d)$ generate the same topology on V . □

In particular, we may consider the seminorms

$$\rho_n(x) = |x_n|,$$

and $\mathcal{N} = \{\rho_n : n \geq 0\}$. Then, we claim that $\mathcal{T} = \mathcal{T}(\mathcal{N})$.

Proof. A basic open neighborhood of $x \in V$ in $\mathcal{T}(\mathcal{N})$ is given by the sets

$$U(x, F, \varepsilon) = \bigcap_{j \in F} \{(y_n)_n : |x_j - y_j| < \varepsilon\}.$$

Meanwhile, a basic open neighborhood of x in $\mathcal{T}$ is given by the sets

$$U'(x, F, \varepsilon) = \bigcap_{j \in F} \pi_j^{-1}((x_j - \varepsilon, x_j + \varepsilon)).$$

In fact, these are the same subsets of V by the definition of the projection map $\pi_j : V \rightarrow \mathbb{R}$ taking $(x_n)_n \mapsto x_j$.

Therefore, if we have $y \in U(x, F, \varepsilon)$ in $\mathcal{T}(\mathcal{N})$, then if we define

$$\delta = \frac{1}{2} \min\{|x_j - y_j| : y_j \in F\},$$

we have that $U'(y, F, \delta) \subseteq U(x, F, \varepsilon)$. Similarly, if $y \in U'(x, F, \varepsilon)$, then setting δ to equal the same as above, we have $U(y, F, \varepsilon) \subseteq U'(x, F, \varepsilon)$.

This gives $\mathcal{T} = \mathcal{T}(\mathcal{N})$. □

In particular, V is not a normed space.

- (v) Consider $c_{00}(\mathbb{N}, \mathbb{K}) \cong \mathbb{K}[x]$ to consist of all the finitely supported sequences $(x_n)_{n \geq 0}$. This is a vector space. Consider the seminorms $\mathcal{N} = \text{SN}(V)$, and the locally convex space $(V, \mathcal{T}(\mathcal{N}))$. Then, this is a locally convex space that is not metrizable. The topology $\mathcal{T}(\mathcal{N})$ is known as the finest locally convex topology.

General Properties of Topological Vector Spaces

Definition: Let $(V, \mathcal{T})$ be a $\mathbb{K}$ -TVS, and let $\tau \in \text{SN}(V)$. Then, τ is a *continuous seminorm* for $(V, \mathcal{T})$ if $\tau : V \rightarrow [0, \infty) \subseteq \mathbb{R}$ is continuous.

Proposition: Let V be a $\mathbb{K}$ -vector space, $\mathcal{N} \subseteq \text{SN}(V)$ define a locally convex topology on V . If $\tau \in \text{SN}(V)$, then τ is continuous for $(V, \mathcal{T}(\mathcal{N}))$ if and only if there exists $C \geq 0$, $n \in \mathbb{N}$, and $\rho_1, \dots, \rho_n \in \mathcal{N}$ such that

$$\tau(v) \leq C \max\{\rho_j(v) : j = 1, \dots, n\} \tag{*}$$

or

$$\tau(v) \leq C \sum_{j=1}^n \rho_j(v).$$

Proof. Suppose τ is continuous. Then, $[0, 1)$ is open in $[0, \infty)$, so $\tau^{-1}([0, 1))$ is open in $\mathcal{T}(\mathcal{N})$ with $0 \in \tau^{-1}([0, 1))$.

So, by definition, there exists $\varepsilon > 0$ and $\rho_1, \dots, \rho_n$ such that

$$\bigcap_{j=1}^n U(0, \rho_j, \varepsilon) \subseteq \tau^{-1}([0, \infty)).$$

That is, for any $i \in [n]$, we have that if $\rho_i(x) < \varepsilon$, then $\tau(x) < 1$.

Let $x \in V$ and $\lambda > 0$. Then,

$$\begin{aligned} \tau(x) &= \tau(\lambda \lambda^{-1} x) \\ &= \lambda \tau(\lambda^{-1} x) \\ &< \lambda \end{aligned}$$

if for every i , we have $\rho_i(\lambda^{-1} x) < \varepsilon$.

Define $J = \{\lambda > 0 : \rho_i(\lambda^{-1} x) < \varepsilon \text{ for all } i \in [n]\}$. Rewriting, we have

$$\begin{aligned} J &= \left\{ \lambda > 0 : \frac{1}{\varepsilon} \rho_i(x) < \lambda \text{ for all } i \in [n] \right\} \\ &= \bigcap_{i=1}^n \left(\frac{1}{\varepsilon} \rho_i(x), \infty \right), \end{aligned}$$

giving $\inf(J) = \frac{1}{\varepsilon} \max\{\rho_i(x) : 1 \leq i \leq n\}$. Therefore, if $\lambda \in J$, then $\tau(x) \leq \lambda$, whence $\tau(x) \leq \inf(J)$. Therefore, we have shown that $\tau(x) \leq \frac{1}{\varepsilon} \max\{\rho_i(x) : 1 \leq i \leq n\}$.

In the reverse direction, suppose $(*)$ holds. Let $x \in V$, $\rho_1, \dots, \rho_n \in \mathcal{N}$, and let $\eta > 0$. Let

$$y \in \bigcap_{i=1}^n U(x, \rho_i, \eta).$$

By the reverse triangle inequality, we have

$$\begin{aligned} |\tau(y) - \tau(x)| &\leq \tau(y - x) \\ &< \varepsilon, \end{aligned}$$

but

$$\begin{aligned} \tau(y - x) &\leq C \max\{\rho_i(y - x) : 1 \leq i \leq n\} \\ &\leq C\eta \\ &< \varepsilon \end{aligned}$$

if η is small enough. Thus, we have for all $y \in \bigcap_{j=1}^n U(x, \rho_j, \eta)$, then $|\tau(y) - \tau(x)| < \varepsilon$, whence τ is continuous at x . $\square$

Remark: If $\tau \in \mathcal{N}$, then τ is continuous with respect to $\mathcal{T}(\mathcal{N})$.

Proposition: Let $(V, \mathcal{T}(\mathcal{N}))$ be a locally convex topological vector space. Then, $U \subseteq V$ is open if and only if for all $x \in U$, there is some continuous seminorm τ and some $\varepsilon > 0$ such that $U(x, \tau, \varepsilon) \subseteq U$.

Theorem (Continuity of Multilinear Maps on locally convex topological vector spaces): Suppose $n \geq 1$, and $V_1, \dots, V_n, W$ are $\mathbb{K}$ -locally convex topological vector spaces, and let $M: V_1 \times \dots \times V_n \rightarrow W$ be a $\mathbb{K}$ -multilinear map.

Then, M is (jointly) continuous if and only if for all continuous seminorm τ on W , there exist $\rho_1, \dots, \rho_n$ con-

tinuous seminorms on $V_1, \dots, V_n$ respectively such that for all tuples $(x_1, \dots, x_n) \in V_1 \times \dots \times V_n$,

$$\tau(M(x_1, \dots, x_n)) \leq \rho(x_1) \cdots \rho(x_n). \quad (\dagger)$$

Proof. Suppose M is (jointly) continuous. That is, M is equipped with the product topology. Since τ is a continuous seminorm, we have that τ is a continuous map from $W \rightarrow [0, \infty)$, we have $\tau^{-1}([0, 1))$ is open in W . Moreover, $M(0, \dots, 0) = 0 \in \tau^{-1}([0, 1))$. Since M is continuous at $(0, \dots, 0)$, there is a neighborhood $U_1 \times \dots \times U_n$ in $V_1 \times \dots \times V_n$ where each U_i is a neighborhood of 0 in V_i , such that $M(U_1 \times \dots \times U_n) \subseteq \tau^{-1}([0, 1))$.

Let $(x_1, \dots, x_n) \in V_1 \times \dots \times V_n$, and let $\lambda_1, \dots, \lambda_n > 0$. We have

$$\begin{aligned} \tau(M(x_1, \dots, x_n)) &= \tau(M(\lambda_1 \lambda_1^{-1} x_1, \dots, \lambda_n \lambda_n^{-1} x_n)) \\ &= \lambda_1 \cdots \lambda_n \tau(M(\lambda_1^{-1} x_1, \dots, \lambda_n^{-1} x_n)) \\ &< \lambda_1 \cdots \lambda_n \end{aligned}$$

if for all i , we have $\lambda_i^{-1} x_i \in U_i$. For each $i \in [n]$, we have U_i is an open neighborhood of $0 \in V_i$, so there exists some continuous seminorm σ_i and $\varepsilon_i > 0$ such that $U(0, \sigma_i, \varepsilon_i) \subseteq U_i$. If for every i , we have $\sigma_i(\lambda_i^{-1} x_i) < \varepsilon_i$, then

$$\tau(M(x_1, \dots, x_n)) < \lambda_1 \cdots \lambda_n.$$

Rewriting, we have $\lambda_i > \frac{1}{\varepsilon_i} \sigma_i(x_i)$ for each such i , so that we have

$$\begin{aligned} \tau(M(x_1, \dots, x_n)) &\leq \lambda_1 \cdots \lambda_n \\ &\leq \inf \left\{ \lambda_1 \cdots \lambda_n : \lambda_i > 0 \text{ and for all } i \in [n], \lambda_i > \frac{1}{\varepsilon_i} \sigma_i(x_i) \right\} \\ &= \left(\frac{1}{\varepsilon_1} \sigma_1(x_1) \right) \cdots \left(\frac{1}{\varepsilon_n} \sigma_n(x_n) \right) \end{aligned}$$

Setting $\rho_i = \frac{1}{\varepsilon_i} \sigma_i$ yields the desired result.

Assume that $(\dagger)$ holds. Let $(x_1, \dots, x_n), (y_1, \dots, y_n) \in V_1 \times \dots \times V_n$. We want to try to compare the difference between $M(x_1, \dots, x_n)$ and $M(y_1, \dots, y_n)$. From multilinearity, we have

$$\begin{aligned} M(x_1, \dots, x_n) - M(y_1, \dots, y_n) &= \sum_{i=1}^n (M(y_1, \dots, y_{i-1}, y_i, x_{i+1}, \dots, x_n) - M(y_1, \dots, y_{i-1}, x_i, x_{i+1}, \dots, x_n)) \\ &= \sum_{i=1}^n M(y_1, \dots, y_{i-1}, y_i - x_i, x_{i+1}, \dots, x_n) \end{aligned}$$

Let $Z \subseteq W$ be open containing $z = M(x_1, \dots, x_n)$, and write $z' = M(y_1, \dots, y_n)$. Then, there exists a continuous seminorm τ on W and $\varepsilon > 0$ such that $U(z, \tau, \varepsilon) \subseteq Z$.

Consider $\tau(z - z')$. Then, we have by multilinearity and the triangle inequality that

$$\tau(z - z') \leq \sum_{i=1}^n \tau(M(y_1, \dots, y_{i-1}, y_i - x_i, x_{i+1}, \dots, x_n)).$$

Using $(\dagger)$ gives continuous seminorms $\rho_1, \dots, \rho_n$ on $V_1, \dots, V_n$ respectively such that

$$\begin{aligned} &\leq \sum_{i=1}^n \rho_1(y_1) \cdots \rho_{i-1}(y_{i-1}) \rho_i(y_i - x_i) \rho_{i+1}(x_{i+1}) \cdots \rho_n(x_n) \\ &\leq n(C + \eta)^{n-1} \eta, \end{aligned}$$

where we assume that for all i , $y_i \in U(x_i, \rho_i, \eta)$, and $C = \max\{\rho_i(x_i) : i = 1, \dots, n\}$. By choosing η sufficiently small, $n(C + \eta)^{n-1}\eta < \varepsilon$, which yields

$$M\left(\prod_{i=1}^n U(x_i, \rho_i, \eta)\right) \subseteq U(z, \tau, \varepsilon) \subseteq Z,$$

so M is continuous. $\square$

Theorem (Refinement of Previous Theorem): Let $V_1, \dots, V_n$ and W be locally convex topological vector spaces, and let $\mathcal{N}$ be a defining collection of seminorms for the topology on W . Then, if $M: V_1 \times \dots \times V_n \rightarrow W$ is a multilinear map, we have M is continuous if and only if for every $\tau \in \mathcal{N}$, there exist continuous seminorms $\rho_1, \dots, \rho_n$ on $V_1, \dots, V_n$ respectively such that for all $x = (x_1, \dots, x_n)$,

$$\tau(M(x)) \leq \prod_{i=1}^n \rho_i(x_i).$$

Proof. The forward direction follows from the fact that τ is a continuous seminorm, while the reverse direction follows from the fact that any continuous seminorm is dominated by a constant multiple of some family of seminorms in $\mathcal{N}$, and a constant multiple of the continuous seminorm ρ_i on V_i is also a continuous seminorm on V_i . $\square$

Corollary: If $T: V \rightarrow W$ is a linear map of locally convex topological vector spaces $\mathcal{N}$ defining the topology on W , then T is continuous if and only if for all $\tau \in \mathcal{N}$ (or all continuous seminorms on W), there exists a continuous seminorm ρ on V such that for all $x \in V$, $\tau(T(x)) \leq \rho(x)$.

Proposition: Let V be a $\mathbb{K}$ -vector space, and let $\mathcal{N}_1, \mathcal{N}_2 \subseteq \text{SN}(V)$. Then, $\mathcal{T}(\mathcal{N}_1) \subseteq \mathcal{T}(\mathcal{N}_2)$ if and only if for all $\tau \in \mathcal{N}_1$, τ is a continuous seminorm for the locally convex topological vector space $(V, \mathcal{T}(\mathcal{N}_2))$.

Proof. We have that $\mathcal{T}(\mathcal{N}_1)$ is contained in $\mathcal{T}(\mathcal{N}_2)$ if and only if the identity map $\text{id}: (V, \mathcal{T}(\mathcal{N}_2)) \rightarrow (V, \mathcal{T}(\mathcal{N}_1))$ is continuous. $\square$

Remark: We recover the fact that $(V, \|\cdot\|_1)$ and $(V, \|\cdot\|_2)$ are equivalent as norms if and only if the identity map $\text{id}: (V, \|\cdot\|_1) \rightarrow (V, \|\cdot\|_2)$ is a homeomorphism.

Normability and Metrizability

We will focus on Hausdorff locally convex topological vector spaces from now on. That is, the defining family of seminorms separates points.

Definition: Let $(V, \mathcal{T}(\mathcal{N}))$ be a Hausdorff locally convex topological vector space. We say V is normable if and only if there exists a norm $\|\cdot\|$ on V such that $\mathcal{T}(\mathcal{N}) = \mathcal{T}(\|\cdot\|)$.

Theorem: Let $(V, \mathcal{T}(\mathcal{N}))$ be a locally convex topological vector space. The following are equivalent:

- (i) $(V, \mathcal{T}(\mathcal{N}))$ is normable;
- (ii) there exists a finite subset F of $\mathcal{T}(\mathcal{N})$ such that $\mathcal{T}(\mathcal{N}) = \mathcal{T}(F)$;

Proof. We start by showing that (i) implies (ii). Let $\|\cdot\|$ be a norm such that $\mathcal{T}(\mathcal{N}) = \mathcal{T}(\|\cdot\|)$. Since $\mathcal{T}(\|\cdot\|) \subseteq \mathcal{T}(\mathcal{N})$, there exist $\rho_1, \dots, \rho_n \in \mathcal{N}$ and some constant C such that

$$\|\cdot\| \leq C \max\{\rho_k : k = 1, \dots, n\}.$$

Let $\tau \in \mathcal{N}$. Then, since $\mathcal{T}(\mathcal{N}) \subseteq \mathcal{T}(\|\cdot\|)$, it follows that there is some K such that $\tau \leq K\|\cdot\|$, whence

$$\tau \leq CK \max\{\rho_k : k = 1, \dots, n\}$$

In particular, this means that the finite subset of seminorms $F = \{\rho_1, \dots, \rho_n\}$ generates the topology $\mathcal{T}(\mathcal{N})$ since every seminorm in $\mathcal{N}$ is continuous with respect to $\mathcal{T}(F)$.

Next, we show that (ii) implies (i). Suppose there is a finite subset $F \subseteq \mathcal{N}$ with elements $\rho_1, \dots, \rho_n$ such

that $\mathcal{T}(\mathcal{N}) = \mathcal{T}(\mathcal{F})$. In particular, we have that for all seminorms $\tau \in \mathcal{N}$, we have

$$\tau \leq C \max\{\rho_k : k = 1, \dots, n\}.$$

We claim that

$$\|\cdot\| = \max\{\rho_k : k = 1, \dots, n\}$$

induces the topology on $\mathcal{N}$. This follows from the fact that every seminorm $\tau \in \mathcal{N}$ is continuous with respect to $\|\cdot\|$, so $\mathcal{T}(\mathcal{N}) \subseteq \mathcal{T}(\|\cdot\|) \subseteq \mathcal{T}(\mathcal{N})$, with the latter inclusion emerging from the fact that $\|\cdot\|$ is necessarily a continuous seminorm in $\mathcal{N}$ by the way it was defined. Since $(V, \mathcal{T}(\mathcal{N}))$ is Hausdorff, it follows that $\|\cdot\|$ separates points, so $\|\cdot\|$ is a norm. $\square$

This allows us to prove that certain examples we discussed earlier are not normable spaces.

Example: Consider

$$V = \mathbb{R}^{\mathbb{N}}$$

with topology defined by seminorms $\rho_n = |x_n|$. Then, we observe that there are no finite subsets of the seminorms $\mathcal{N} = \{\rho_k : k \in \mathbb{N}\}$ that separate the points of V by virtue of the fact that if $F = \{\rho_{i_1}, \dots, \rho_{i_j}\}$ is a finite subset, then the elementary vector e_{i_j+1} with 1 in position $(i_j + 1)$ is taken to 0 by any continuous seminorm on $\mathcal{T}(\mathcal{F})$, but e_{i_j+1} is not zero.

Next, we concern ourselves with metrizability of topological vector spaces.

Recall that (X, τ) is *first countable* if for any $x \in X$, the system of open neighborhoods $\mathcal{O}_x$ has a countable basis (i.e., a countable neighborhood basis). That is, there is some $\{U_n : n \geq 1\} \subseteq \mathcal{O}_x$ such that for any $Z \in \mathcal{O}_x$, there is some n such that $U_n \subseteq Z$. Metric spaces are always first countable by virtue of the Archimedean property.

Proposition: A (not necessarily locally convex) TVS $(V, \mathcal{T})$ is first countable if and only if 0 has a countable neighborhood basis.

Proof. Addition by a fixed vector $v \in V$ is a homeomorphism of V . $\square$

Theorem: Let $(V, \mathcal{T}(\mathcal{N}))$ be a locally convex topological vector space. The following are equivalent:

- (i) the space $(V, \mathcal{T}(\mathcal{N}))$ is metrizable;
- (ii) the space $(V, \mathcal{T}(\mathcal{N}))$ is first countable;
- (iii) there exists a countable subset $\mathcal{M} \subseteq \mathcal{N}$ such that $\mathcal{T}(\mathcal{M}) = \mathcal{T}(\mathcal{N})$.

Proof. We see that (i) implies (ii) since metrizable spaces are first countable.

Next, we show that (ii) implies (iii). Let $\{U_n : n \geq 0\}$ be a neighborhood basis for 0. Then, for any n , there exists finite $F_n \subseteq \mathcal{N}$ and $\varepsilon_n > 0$ such that

$$\bigcap_{\rho \in F_n} U(0, \rho, \varepsilon_n) \subseteq U_n.$$

Taking $\mathcal{M} = \bigcup_{n \geq 0} F_n$, this yields a countable subcollection of $\mathcal{N}$. We will show that $\mathcal{T}(\mathcal{N}) \subseteq \mathcal{T}(\mathcal{M})$. That is, we will have to show that for all $\tau \in \mathcal{N}$, τ is continuous with respect to $\mathcal{M}$.

We have $\tau^{-1}([0, 1])$ is a neighborhood of 0 in $\mathcal{T}(\mathcal{N})$. This contains U_n for some $n \in \mathbb{N}$, and thus some finite subset F_n with

$$\begin{aligned} \bigcap_{\rho \in F_n} U(0, \rho, \varepsilon_n) &\subseteq U_n \\ &\subseteq \tau^{-1}([0, 1]). \end{aligned}$$

In particular, this means that if $x \in V$ is such that for all $\rho \in F_n$, $\rho(x) < \varepsilon_n$, then $x \in \tau^{-1}([0, 1))$, whence τ is continuous. Scaling allows one to apply the same method for any $x \in V$.

Finally, we will show that (iii) implies (i). Let $\mathcal{M} \subseteq \mathcal{N}$ be a countable subset with seminorms $\{\rho_n : n \geq 0\}$. Let

$$d(x, y) = \sum_{n=0}^{\infty} 2^{-n} \min\{\rho_n(x - y), 1\}.$$

Then, this is a distance metric and $T(d) = T(\mathcal{M})$. □

Definition: A locally convex topological vector space (V, T) . If (V, T) is complete metrizable, then we call V a *Fréchet space*.

Example:

- (i) Banach spaces are Fréchet spaces.
- (ii) The space of rapidly decreasing sequences $s(\mathbb{N}, \mathbb{K})$ is a Fréchet space that is not normable. The distance metric is given by

$$d(x, y) = \sum_{k=0}^{\infty} 2^{-k} \min(1, \rho_k(x - y)).$$

- (iii) The vector space $V = \mathbb{R}^{\mathbb{N}}$ with the product topology is a Fréchet space.
- (iv) The space of finitely supported sequences $c_{00}(\mathbb{N}, \mathbb{K})$, henceforth denoted c_{00} , with the finest locally convex topology is a topological vector space that is not metrizable, let alone a Fréchet space. This follows from a diagonal argument on the assumption of countably many seminorms.

Consider the uncountable collection of seminorms $\mathcal{N}$ given by, for any $\omega \in [0, \infty)^{\mathbb{N}}$,

$$\rho_{\omega}(x) = \sum_{n=0}^{\infty} \omega_n |x_n|,$$

where we observe that each ρ_{ω} is distinct. It is possible to show that $T(\mathcal{N}) = T(\text{SN}(c_{00}))$.

One direction follows directly from the definition. The reverse direction uses the fact that $\{e_n : n \geq 0\}$ forms an algebraic basis for c_{00} . If ρ is a seminorm, then we will define $\omega_n = \rho(e_n)$. For any $x \in c_{00}$, then we may write

$$x = \sum_{n=0}^N x_n e_n,$$

satisfying

$$\begin{aligned} \rho(x) &= \rho\left(\sum_{n=0}^N x_n e_n\right) \\ &\leq \sum_{n=0}^N \omega_n |x_n| \\ &= \rho_{\omega}(x). \end{aligned}$$

Since x is arbitrary, it follows that every seminorm is continuous with respect to the seminorms of the family $\mathcal{N}$, whence the family $\mathcal{N}$ defines the topology on c_{00} .

In fact, we may assume that $\omega \in \mathbb{N}^{\mathbb{N}}$ by replacing $(\omega_n)_n \in [0, \infty)^{\mathbb{N}}$ with $(\lceil \omega_n \rceil)_n$.

Notable Examples

Example: Let $K \subseteq \mathbb{R}^d$ be a compact subset. Define

$$\mathcal{D}_K(\mathbb{R}^d) = C_K^\infty(\mathbb{R}^d)$$

to be the space of C^∞ with $\text{supp}(f) \subseteq K$. Define seminorms for each $N \geq 0$ by

$$\rho_N(f) = \max_{\substack{\alpha \in \mathbb{N}^d \\ |\alpha| \leq N}} \sup_{x \in \mathbb{R}^d} |\partial^\alpha f(x)|.$$

Let $\mathcal{T} = \mathcal{T}(\{\rho_N : N \geq 0\})$. This is a countable collection of seminorms, so the space $C_K^\infty(\mathbb{R}^d)$ equipped with $\mathcal{T}$ is a Fréchet space with the metric

$$d(f, g) = \sum_{N \geq 0} 2^{-N} \min(1, \rho_N(f - g)).$$

Example: A function $f: \mathbb{R}^d \rightarrow \mathbb{C}$ is called a Schwartz function if $f \in C^\infty(\mathbb{R}^d)$ and for all $\alpha \in \mathbb{N}^d$ and all $k \in \mathbb{N}$,

$$\sup_{x \in \mathbb{R}^d} \langle x \rangle^k |\partial^\alpha f(x)| < \infty.$$

We label these seminorms by $\rho_{\alpha, k}(f)$. The space $\mathcal{S}(\mathbb{R}^d)$ of all such functions is known as the *Schwartz space* for $\mathbb{R}^d$.

The standard topology on $\mathcal{S}(\mathbb{R}^d)$ is given by $\mathcal{T} = \mathcal{T}(\mathcal{N})$, where $\mathcal{N}$ is the collection of all such $\rho_{\alpha, k}$.

We can also consider the family of all such ρ_N with

$$\rho_N(f) = \max_{\substack{|\alpha| \leq N \\ k \leq N}} \rho_{\alpha, k}(f),$$

which yields the metric

$$d(f, g) = \sum_{N=0}^{\infty} 2^{-N} \min(1, \rho_N(f - g)).$$

This induces the same topology as the family of all the $\rho_{\alpha, k}$.

With this distance metric, the Schwartz space is complete. In particular, this means that $\mathcal{S}(\mathbb{R}^d)$ is an example of a Fréchet space. The Schwartz space is needed to define the Fourier transform (which we will discuss more heavily later). The Fourier transform is defined by

$$\hat{f}(\xi) = \int_{\mathbb{R}^d} e^{-ix\xi} f(x) d^d x.$$

When $f \in L^1$, we have that $\hat{f}$ is well-defined, but $\hat{f}$ is not necessarily in L^1 . It turns out that the Fourier transform is an automorphism of the Schwartz space (over $\mathbb{C}$). Note that Schwartz functions are L^1 since they decay faster than functions of the form

$$u(x) = \langle x \rangle^\lambda$$

where $\lambda < d$. That such a $u(x)$ is integrable can be found by using the hyperspherical coordinates.

Example (A Non-Example): The dual space of the Schwartz space, $\mathcal{S}(\mathbb{R}^d)^*$, is called the space of *tempered*

distributions. We say $\mathcal{S}(\mathbb{R}^d)^* \subseteq C_c^\infty(\mathbb{R}^d)^*$, where the dual space $C_c^\infty(\mathbb{R}^d)^*$ is called the space of all distributions. We will discuss these more soon.

The space $C_c(\mathbb{R}^d)$ is the space of compactly supported C^∞ functions. In particular,

$$\begin{aligned} C_c^\infty(\mathbb{R}^d) &= \bigcup_{K \text{ compact}} C_K^\infty(\mathbb{R}^d) \\ &= \bigcup_{N \geq 1} C_{B(0,N)}^\infty(\mathbb{R}^d), \end{aligned}$$

where

$$B(0, N) = \{x \in \mathbb{R}^d : |x| \leq N\}$$

is the closed ball centered at 0.

Though $C_c^\infty(\mathbb{R}^d)$ is the increasing countable union of Fréchet spaces, it is not itself Fréchet; in fact, $C_c^\infty(\mathbb{R}^d)$ is not even metrizable. It is an example of an inductive limit of a family of Fréchet spaces.

Let

$$\iota_K : C_K^\infty(\mathbb{R}^d) \rightarrow C_c^\infty(\mathbb{R}^d)$$

denote the inclusion maps. We say a seminorm ρ on $C_c^\infty(\mathbb{R}^d)$ is compatible if and only if for every compact $K \subseteq \mathbb{R}^d$, $\rho \circ \iota_K$ is a continuous seminorm on $C_K^\infty(\mathbb{R}^d)$. The topology on $C_c^\infty(\mathbb{R}^d)$ is the topology generated by all compatible seminorms.